

Tensor Network Representations of Discrete Maximum Entropy Distributions via Mean Polytopes

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

The Shannon entropy of a distribution quantifies the amount of structure, or information, and appears therefore as an important concept across many scientific areas. By Shannons source code theorem [1], entropy describes fundamental limits of information transfer. In statistical physics, founded on the ergodic assumptions, maximum entropy distributions such as Maxwell-Boltzmann distributions are the gold standard to model equilibrium distributions.

The principle of entropy maximization also appears naturally in learning problems. Extracting characteristics observed in data does not determine an estimated model yet. Choosing the distribution with the least structure among those reflecting the observed characteristics is a regularization principle. Here one can choose the Shannon entropy to quantify the negative amount of structure in the distribution and maximize

the quantity with respect to the constraints. These maximum entropy problems are the dual problems to convex maximum likelihood problems [2]. Maximum entropy principles appear further in reinforcement learning [3, 4], AGI [5] and Bayesian NN [6].

A classical result states that maximum entropy distributions are in exponential families, if and only if the corresponding mean parameter is in the relative interior of the mean polytope [7]. A natural extension is therefore the closure of the exponential families as investigated in [8, 9]. Alternative methods use toric ideals to allow for infinite weight vectors [10]. Numerical stability of the maximum entropy distributions near the boundary is studied in [11, 12].

Less obvious, studying general maximum entropy distributions might be a key step towards a mathematically sound unification of logical and probabilistic models. Combining logical and probabilistic AI is a necessity in statistical relational AI, neuro-symbolic AI and explainable AI [13–15]. Probability theory with exponential families typically focuses on distributions with maximal support [2]. Logics on the contrary can be understood as a study of the support of models, marking the worlds consistent with a knowledge base. Logical and probabilistic models can be treated in the tensor network formalism, which thus serves as a unifying representation language [16].

We in this work derive tensor network representations of all discrete maximum entropy distributions, based on the framework presented in [16, 17]. Tensor network formats originate from quantum many-body physics [18] as a mitigation scheme for the curse of dimensionality [19]. From a numerical perspective [20, 21] formats like the CP format [22, 23] and the TT format [24] have been studied. They have been applied to various areas such as in high-dimensional PDEs [25, 26], or quantum simulation [27]. Lastly, tensor network schemes are applied to artificial intelligence [28–30].

1.1 Tensor Notation

We introduce tensors based on index variables X_k to each of the $k \in [d] = \{0, \dots, d-1\}$ axes, each taking values in $[m_k] = \{0, \dots, m_k - 1\}$, where $m_k \in \mathbb{N}$ is the dimension of the variable. Tensors are then maps

$$\tau [X_{[d]}] : \prod_{k \in [d]} [m_k] \rightarrow \mathbb{R}$$

from indices $x_{[d]} = (x_0, \dots, x_{d-1}) \in \prod_{k \in [d]} [m_k]$ to their coordinates $\tau [X_{[d]} = x_{[d]}] \in \mathbb{R}$. One-hot encodings of index tuples $x_{[d]}$ are specific tensors $\epsilon_{x_{[d]}} [X_{[d]}]$ such that $\epsilon_{x_{[d]}} [X_{[d]} = \tilde{x}_{[d]}] = 1$ if $\tilde{x}_{[d]} = x_{[d]}$ and 0 else.

Given a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of an arbitrary node set $\mathcal{V}$ and hyperedges $e \in \mathcal{E}$ being subsets of $\mathcal{V}$, we define decorating tensor network as follows. Each node $v \in \mathcal{V}$ gets decorated by a variable X_v of dimension m_v and each hyperedge $e \in \mathcal{E}$ by a tensor $\tau^e [X_e]$ with the index variables to the contained nodes. The set of decorating tensors then builds the tensor network $\tau^{\mathcal{G}}$ on the hypergraph. A contraction of the tensor network keeping the nodes $\mathcal{U} \subset \mathcal{V}$ open is then the tensor $\langle \tau^{\mathcal{G}} \rangle_{[X_{\mathcal{U}}]}$ with variables $X_{\mathcal{U}}$

and coordinates to the indices $x_{\mathcal{U}}$ by

$$\langle \tau^{\mathcal{G}} \rangle_{[X_{\mathcal{U}}=x_{\mathcal{U}}]} = \sum_{x_{\mathcal{V} \setminus \mathcal{U}}} \prod_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

1.2 The Maximum Entropy Problem

Given a non-negative and non-vanishing base measure $\nu [X_{[d]}]$, a probability distribution is a non-negative tensor $\mathbb{P} [X_{[d]}]$ such that

$$\langle \mathbb{P} [X_{[d]}] , \nu [X_{[d]}] \rangle_{[\emptyset]} = 1 .$$

We denote the set of such distributions by

$$\Lambda^{\nu} = \left\{ \mathbb{P} [X_{[d]}] : \mathbb{P} [X_{[d]}] \geq 0 [X_{[d]}] , \langle \mathbb{P} [X_{[d]}] , \nu [X_{[d]}] \rangle_{[\emptyset]} = 1 , \right. \\ \left. \forall_{x_{[d]} \in \times_{k \in [d]} [m_k]} \nu [X_{[d]} = x_{[d]}] = 0 \Rightarrow \mathbb{P} [X_{[d]} = x_{[d]}] = 0 \right\} .$$

Note that we here restrict the support of the tensor $\mathbb{P} [X_{[d]}]$ to be included in the support of the base measure, to avoid ambiguity in the representation.

The entropy of a distribution relative to the base measure ν is

$$\mathbb{H}_{\nu} [\mathbb{P} [X_{[d]}]] = - \langle \mathbb{P} [X_{[d]}] , \ln [\mathbb{P} [X_{[d]}]] , \nu [X_{[d]}] \rangle_{[\emptyset]} .$$

Note that this quantity depends on the chosen base measure.

The mean parameter of a distribution $\mathbb{P} [X_{[d]}]$ to a statistic $t : \times_{k \in [d]} [m_k] \rightarrow \mathbb{R}^p$ is the vector $\mu [L] \in \mathbb{R}^p$ with the coordinates

$$\mu [L = \ell] = \mathbb{E} [f_{\ell}] = \langle t_{\ell} [X_{[d]}] , \mathbb{P} [X_{[d]}] , \nu [X_{[d]}] \rangle_{[\emptyset]} .$$

To express the computation of the mean parameter by a single contraction we introduce a coordinate selection variable L of dimension p and the selection encoding $\sigma^t [X_{[d]}, L]$ with coordinates by

$$\sigma^t [X_{[d]} = x_{[d]}, L = \ell] = t_{\ell} [X_{[d]} = x_{[d]}] .$$

The mean parameter is then

$$\mu [L] = \langle \mathbb{P} [X_{[d]}] , \sigma^t [X_{[d]}, L] \rangle_{[L]} .$$

The maximum entropy problem given a mean parameter $\mu^* [L]$ is stated by

$$\operatorname{argmax}_{\mathbb{P} [X_{[d]}] \in \Lambda^{\nu}} \mathbb{H}_{\nu} [\mathbb{P} [X_{[d]}]] \quad \text{subject to} \quad \langle \mathbb{P} [X_{[d]}] , \sigma^t [X_{[d]}, L] , \nu [X_{[d]}] \rangle_{[L]} = \mu^* [L] \\ (P_{t, \mu, \nu})$$

1.3 Computation-Activation Networks

We now introduce with Computation-Activation Networks families of tensor networks representing, as we will show in this work, the solution of any instance of Problem $\mathbf{P}_{t,\mu,\nu}$. Given a statistic $t : \times_{k \in [d]} [m_k] \rightarrow \mathbb{R}^p$ we define an index enumeration variable Y_ℓ of dimension $|\text{im}(t_\ell)|$ and injective index enumeration vectors

$$I_\ell [Y_\ell] : [n_\ell] \rightarrow \mathbb{R}.$$

Based on the enumeration of the image we define the basis encoding tensor of the statistic as

$$\beta^t [Y_{[p]}, X_{[d]}] = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \left(\bigotimes_{\ell \in [p]} \epsilon_{I_\ell^{-1}(t_\ell[x_{[d]}])} [Y_\ell] \right) \otimes \epsilon_{x_{[d]}} [X_{[d]}].$$

A computation network is any representation of $\beta^t [Y_{[p]}, X_{[d]}]$ as a tensor network. These can be constructed in the case statistics being a composition of connective functions, see [16].

An activation tensor is $\tau [Y_{[p]}]$ and the corresponding Computation-Activation Network with respect to the statistic t and the base measure $\nu [X_{[d]}]$ the tensor

$$\mathbb{P} [X_{[d]}] = \frac{\langle \beta^t [Y_{[p]}, X_{[d]}], \tau [Y_{[p]}] \rangle_{[X_{[d]}]}}{\langle \beta^t [Y_{[p]}, X_{[d]}], \tau [Y_{[p]}], \nu [X_{[d]}] \rangle_{[X_{[d]}]}}.$$

We are interested in decomposition formats of $\tau [Y_{[p]}]$, where we use sets of tensor networks $\mathcal{T}^\mathcal{G}$ on a hypergraph $\mathcal{G}$.

Definition 1 The Computation-Activation Networks to a statistic t and a hypergraph $\mathcal{G}$ is the family

$$\Lambda^{t,\mathcal{G},\nu} = \left\{ \frac{\langle \tau [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[X_{[d]}]}}{\langle \tau [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]}} : \tau [Y_{[p]}] \in \mathcal{T}^\mathcal{G} \right\}.$$

1.4 Contributions

We exploit the novel tensor network formalism developed in [16] to represent generic maximum entropy distributions based on the position of their mean parameters in the mean polytope, see Thm. 4. The connection to logical formalisms of artificial intelligence is exploited in the study of boolean statistics, where we interpret mean polytopes based on satisfiability of formulas (see Thm. 7) and construct statistics to arbitrary polytopes (see Thm. 8).

1.5 Structure of the paper

To prepare for the presentation of our main results we introduce in section Sect. 2 the mean polytope, its faces, and their tensor network representations. In Sect. 3 we study exponential families as classes of elementary Computation-Activation Networks and show that they are the maximum entropy distributions to mean parameters in the relative interior of the mean polytope. This result is generalized in Sect. 4 where we characterize generic maximum entropy distributions based on Computation-Activation Networks. In Sect. 5 we investigate consequences of these results in case of boolean statistics and connect them to propositional logics.

2 The mean polytope

The mean polytope is the set of possible mean parameters given a by base measure representable distribution in Λ^ν . We define it as

$$\mathcal{M}_{t,\nu} := \left\{ \left\langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} : \mathbb{P}[X_{[d]}] \in \Lambda^\nu \right\},$$

where we denote by Λ^ν the set of all probability distributions representable with respect to ν , and by $\sigma^t[X_{[d]}, L]$ the selection encoding of the statistic (see introduction).

2.1 Convex hull

First of all, we characterize the mean polytope as the convex hull of the slices to the selection encoding.

Lemma 1 For any statistic t and base measure ν , the set of mean parameters is the convex hull of the set

$$\mathcal{N}_{t,\nu} := \{ \sigma^t[X_{[d]} = x_{[d]}, L] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \}$$

that is $\mathcal{M}_{t,\nu} = \text{conv}(\mathcal{N}_{t,\nu})$.

Proof This follows from

$$\Lambda^\nu = \text{conv} \left(\frac{1}{\nu[X_{[d]} = x_{[d]}]} \cdot \epsilon_{x_{[d]}}[X_{[d]}] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \right).$$

and for any vertex of this simplex we have

$$\left\langle \frac{1}{\nu[X_{[d]} = x_{[d]}]} \cdot \epsilon_{x_{[d]}}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} = \sigma^t[X_{[d]} = x_{[d]}, L].$$

□

2.2 Faces

Let us now continue with the investigation of the faces of the mean parameter polytope.

Definition 2 We say that the convex hull of a subset $\mathcal{N}_{t,\nu}^{\mathcal{F}} \subset \mathcal{N}_{t,\nu}$ is a face of a mean polytope $\mathcal{M}_{t,\nu}$, if and only if there is a face normal vector $\theta_{\mathcal{F}}[L] \in \mathbb{R}^p$ such that

$$\mathcal{N}_{t,\nu}^{\mathcal{F}} = \operatorname{argmax}_{\mu[L] \in \mathcal{N}_{t,\nu}} \langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]} .$$

We denote the face as $\mathcal{F} = \operatorname{conv}(\mathcal{N}_{t,\nu}^{\mathcal{F}})$. The set of all faces of the mean polytope $\mathcal{M}_{t,\nu}$ is denoted by $L(\mathcal{M}_{t,\nu})$.

The set $L(\mathcal{M}_{t,\nu})$, which is partially ordered by the inclusion operation, is a lattice [31]. We notice, that each face itself is a convex polytope. What is more, we can characterize these as mean parameter polytopes with respect to refined base measures to be defined next.

Definition 3 The to a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ refined base measure is the tensor $\nu^{t,\mathcal{F}}[X_{[d]}]$ with coordinates to $x_{[d]} \in \times_{k \in [d]}[m_k]$ by

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \begin{cases} \nu[X_{[d]} = x_{[d]}] & \text{if } t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}} \\ 0 & \text{else} \end{cases} .$$

We now specify the mean parameter polytope to any face using the face measure as a refinement of the base measure.

Lemma 2 For any face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have

$$\mathcal{F} = \mathcal{M}_{t,\nu^{t,\mathcal{F}}} .$$

Proof For any $x_{[d]} \in \times_{k \in [d]}[m_k]$ we have $\nu^{t,\mathcal{F}}[X_{[d]} = x_{[d]}] \neq 0$ if and only if $t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and $\nu[X_{[d]} = x_{[d]}] \neq 0$. Thus we have

$$\begin{aligned} \mathcal{F} &= \operatorname{conv}(\mathcal{N}_{t,\nu}^{\mathcal{F}}) \\ &= \operatorname{conv}(\sigma^t[X_{[d]} = x_{[d]}, L] : t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}, \nu[X_{[d]} = x_{[d]}] \neq 0) \\ &= \operatorname{conv}(\sigma^t[X_{[d]} = x_{[d]}, L] : \nu^{t,\mathcal{F}}[X_{[d]} = x_{[d]}] \neq 0) \\ &= \mathcal{M}_{t,\nu^{t,\mathcal{F}}} . \square \end{aligned}$$

Representability of a distribution with respect to face measures is an equivalent condition for the mean parameter of a distribution to be on a face, as we show next.

Lemma 3 If and only if for a distribution $\mathbb{P} \left[X_{[d]} \right] \in \Lambda^\nu$ and a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have

$$\left\langle \sigma^t \left[X_{[d]}, L \right], \mathbb{P} \left[X_{[d]} \right], \nu \left[X_{[d]} \right] \right\rangle_{[L]} \in \mathcal{F},$$

then

$$\mathbb{P} \left[X_{[d]} \right] \in \Lambda^{\nu^{t,\mathcal{F}}}.$$

Proof For a distribution $\mathbb{P} \left[X_{[d]} \right] \in \Lambda^\nu$ to be in $\Lambda^{\nu^{t,\mathcal{F}}}$, it is enough to show that $\mathbb{P} \left[X_{[d]} \right]$ is only supported at $\nu^{t,\mathcal{F}} \left[X_{[d]} \right]$. We have

$$\mu[L] = \sum_{x_{[d]}} \mathbb{P} \left[X_{[d]} = x_{[d]} \right] \cdot \nu \left[X_{[d]} = x_{[d]} \right] \cdot \gamma^t \left[X_{[d]} = x_{[d]}, L \right].$$

Let now $\theta_{\mathcal{F}}[L]$ be a face normal to the face $\mathcal{F}$. We then have

$$\langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]} = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P} \left[X_{[d]} = x_{[d]} \right] \cdot \nu \left[X_{[d]} = x_{[d]} \right] \cdot \left\langle \gamma^t \left[X_{[d]} = x_{[d]}, L \right], \theta_{\mathcal{F}}[L] \right\rangle_{[\emptyset]}.$$

Now if and only if $\mathbb{P} \left[X_{[d]} = x_{[d]} \right] \cdot \nu \left[X_{[d]} = x_{[d]} \right]$ is supported only for $x_{[d]}$ with $t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ we have that

$$\langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]} = \max_{\mu[L] \in \mathcal{M}_{t,\nu}} \langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]}$$

which is equal to $\mu[L] \in \mathcal{F}$. Thus, if and only if $\mu[L] \in \mathcal{F}$ then $\mathbb{P} \left[X_{[d]} \right]$ is only supported at $x_{[d]}$ in the support of $\nu^{t,\mathcal{F}} \left[X_{[d]} \right]$. $\square$

Let us now investigate tensor network representations of refined base measures, based on the basis encoding β^t of a statistic.

Theorem 1 For any face $\mathcal{F}$ of $\mathcal{M}$ we have

$$\nu^{t,\mathcal{F}} \left[X_{[d]} \right] = \left\langle \beta^t \left[Y_{[p]}, X_{[d]} \right], \kappa^{\mathcal{F}} \left[Y_{[p]} \right], \nu \left[X_{[d]} \right] \right\rangle_{[X_{[d]}]}$$

where

$$\kappa^{\mathcal{F}} \left[Y_{[p]} \right] = \sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_{v[L]} \left[Y_{[p]} \right].$$

Proof For any $v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ the tensor

$$\tau^v \left[X_{[d]} \right] = \left\langle \beta^t \left[Y_{[p]}, X_{[d]} \right], \epsilon_{v[L]} \left[Y_{[p]} \right] \right\rangle_{[X_{[d]}]}$$

is the indicator of the preimage of μ under σ^t . Since preimages of the elements in $\mathcal{N}_{t,\nu}^{\mathcal{F}}$ are disjoint, the support of $\tau^v \left[X_{[d]} \right]$ is disjoint and their sum

$$\sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \tau^v \left[X_{[d]} \right]$$

is the indicator of the preimage of $\mathcal{F}$ under σ^t . The face measure obeys thus

$$\begin{aligned}\nu^{t,\mathcal{F}}[X_{[d]}] &= \left\langle \left(\sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \tau^v[X_{[d]}] \right), \nu[X_{[d]}] \right\rangle_{[X_{[d]}]} \\ &= \sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \left\langle \beta^t[Y_{[p]}, X_{[d]}], \epsilon_{v[L]}[Y_{[p]}], \nu[X_{[d]}] \right\rangle_{[X_{[d]}]} \\ &= \left\langle \beta^t[Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}}[Y_{[p]}], \nu[X_{[d]}] \right\rangle_{[X_{[d]}]} \square\end{aligned}$$

We now investigate the representation of refined base measures by Computation-Activation Networks.

Definition 4 Let $\mathcal{G}$ be a hypergraph which nodes include $[p]$. We say that a face $\mathcal{F}$ is representable by $\mathcal{G}$ if and only if there is a set $\mathcal{U}$ of basis vectors with $\mathcal{U} \cap \mathcal{N}_{t,\nu} = \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and there is a tensor network $\kappa^{\mathcal{G}}$ with respect to the hypergraph $\mathcal{G}$ such that

$$\left\langle \kappa^{\mathcal{G}} \right\rangle_{[Y_{[p]}]} = \sum_{v \in \mathcal{U}} \epsilon_v[Y_{[p]}].$$

We call any such tensor network a face activating tensor network.

Lemma 4 If any only if a face is representable by a hypergraph $\mathcal{G}$ we have

$$\frac{1}{\left\langle \nu^{t,\mathcal{F}}[X_{[d]}] \right\rangle_{[\emptyset]}} \cdot \nu^{t,\mathcal{F}}[X_{[d]}] \in \Lambda^{t,\mathcal{G},\nu}.$$

Proof For any $\mathcal{U}$ with $\mathcal{U} \cap \mathcal{N}_{t,\nu} = \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and tensor network $\kappa^{\mathcal{G}}$ respecting the assumptions of Def. 4 we have

$$\begin{aligned}\left\langle \left\langle \kappa^{\mathcal{G}} \right\rangle_{[Y_{[p]}]}, \beta^t[Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]} &= \left\langle \sum_{v \in \mathcal{U}} \epsilon_v[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]} \\ &= \left\langle \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]}.\end{aligned}$$

Here we used in the second equation that only the vertices in $\mathcal{N}_{t,\nu}$ are in the image of t . It follows, that

$$\frac{1}{\left\langle \nu^{t,\mathcal{F}}[X_{[d]}] \right\rangle_{[\emptyset]}} \cdot \nu^{t,\mathcal{F}}[X_{[d]}] = \frac{\left\langle \left\langle \kappa^{\mathcal{G}} \right\rangle_{[Y_{[p]}]}, \beta^t[Y_{[p]}, X_{[d]}] \right\rangle_{[X_{[d]}]}}{\left\langle \left\langle \kappa^{\mathcal{G}} \right\rangle_{[Y_{[p]}]}, \beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}] \right\rangle_{[\emptyset]}}$$

and thus $\nu^{t,\mathcal{F}}[X_{[d]}|\emptyset] \in \Lambda^{t,\mathcal{G},\nu}$. $\square$

Example 1 (Vertices) Vertices $\mathcal{F} = \{v[L]\}$ are proper faces of affine dimension 0, that is they consist in single vectors $v[L]$. Since each vertex is in the image $\mathcal{N}_{t,\nu}$, there exists an index tuple $x_{[d]} \in \mathcal{X}$ such that $\nu[X_{[d]} = x_{[d]}] = 1$ and

$$v[L] = \sigma^t[X_{[d]} = x_{[d]}, L] .$$

Then $\kappa^{\mathcal{F}}[Y_{[p]}]$ is the one-hot encoding of the vector $\sigma^t[X_{[d]} = x_{[d]}, L]$, that is

$$\kappa^{\mathcal{F}}[Y_{[p]}] = \epsilon_{\sigma^t[X_{[d]} = x_{[d]}, L]}[Y_{[p]}] .$$

Such tensors are examples of elementary tensors corresponding to the elementary hypergraph

$$\text{EL} = ([p], \{\{\ell\} : \ell \in [p]\}) .$$

In particular, the activation core is elementary and the corresponding refined base measure is in $\Lambda^{t, \text{EL}, \nu}$.

While vertices are the minimal non-vanishing faces in the face-lattice (see [31]), we now show that also the maximal face, namely the polytope itself, is representable with respect to the elementary hypergraph EL.

Example 2 (Maximal face) The maximal face $\mathcal{F}^\emptyset = \mathcal{M}_{t,\nu}$ coincides with the mean polytope itself and is given by the choice $\theta_\emptyset[L] = 0[L]$. In this case the corresponding activation tensor to the face measure is

$$\kappa^\emptyset[Y_{[p]}] = \mathbb{I}[Y_{[p]}] = \bigotimes_{\ell \in [p]} \mathbb{I}[Y_\ell] ,$$

where by $\mathbb{I}[Y_{[p]}]$ we denote tensors with all coordinates 1. $\kappa^\emptyset Y_{[p]}$ is elementary and the maximal face is always representable by an elementary activation.

2.3 Representation in the CP format

Extending Example 1 and Example 2, we can provide a coarse estimation of the hypergraph $\mathcal{G}$ required to decompose $\kappa^{\mathcal{F}}$ for generic faces $\mathcal{F}$. To this end, we use the CP format describing tensor networks on the CP hypergraph (see Figure 1)

$$\text{CP} = ([p] \cup \{I\}, \{\{\ell, I\} : \ell \in [p]\}) .$$

The dimension of the variable I is called the hidden rank of the format.

Lemma 5 Any face $\mathcal{F}$ is representable by a CP graph with hidden rank

$$r \leq \min \left(|\mathcal{N}_{t,\nu}^{\mathcal{F}}|, |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1 \right) .$$

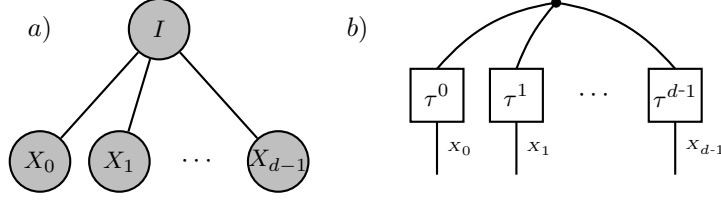


Fig. 1 The CP tensor format consist in the CP hypergraph a) and decorating tensor networks b). Tensors in this format constructed as in Lem. 5 represent arbitrary faces.

Proof We show the claim by constructing two face activating tensor networks to $\mathcal{F}$ in a CP hypergraph with hidden rank $|\mathcal{N}_{t,\nu}^{\mathcal{F}}|$ and in a CP hypergraph with hidden rank $|\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1$. To show the first representation we enumerate the vertices by a variable I with dimension $r = |\mathcal{N}_{t,\nu}^{\mathcal{F}}|$, i.e. $\mathcal{N}_{t,\nu}^{\mathcal{F}} = \{v^i[L] : i \in [n]\}$. Then we define for $\ell \in [p]$ core tensors $\tau^\ell[Y_\ell, I]$

$$\tau^\ell[Y_\ell, I = i] = \epsilon_{v^i[L=\ell]}[Y_\ell] .$$

Then we have

$$\left\langle \{\tau^\ell[Y_\ell, I] : \ell \in [p]\} \right\rangle_{[Y_{[p]}]} = \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}]$$

and thus have found an activation tensor network in a CP graph with hidden rank $|\mathcal{N}_{t,\nu}^{\mathcal{F}}|$ representing the face $\mathcal{F}$.

We continue with the second representation, for which we enumerate the set $\mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^{\mathcal{F}}$ by $v^i[L]$ where $i \in [|\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}|]$. We define variable I with dimension $r = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1$ and define for $\ell \in [p]$ core tensors

$$\tau^\ell[Y_\ell, I = i] = \begin{cases} -\epsilon_{v^i[L=\ell]}[Y_\ell] & \text{if } i < |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| \\ \mathbb{I}[Y_\ell] & \text{if } i = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| \end{cases} .$$

We then have

$$\begin{aligned} \left\langle \{\tau^\ell[Y_\ell, I] : \ell \in [p]\} \right\rangle_{[Y_{[p]}]} &= \mathbb{I}[Y_{[p]}] - \sum_{v \in \mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}] \\ &= \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v[Y_{[p]}] + \sum_{v \in (\times_{\ell \in [p]} [p_\ell])/\mathcal{N}_{t,\nu}} \epsilon_v[Y_{[p]}] . \end{aligned}$$

We have thus found a face activating tensor network for $\mathcal{F}$ in a CP format with hidden rank $r = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^{\mathcal{F}}| + 1$. $\square$

We notice that the CP rank bound of Lem. 5 is tight in Example 1 and Example 2, since the elementary format is a restriction of the CP format to rank 1.

2.4 Examples of elementarily representable faces

We show in the following example, that all faces of a hypercube and a simplex are representable by elementary activation tensors.

Example 3 (Hypercube) In cases where $X_{[d]}$ are boolean and we have $\ell \in [p]$ features

$$f_\ell [X_{[d]} = x_{[d]}] = x_\ell$$

the mean polytope is the hypercube

$$\mathcal{M}_{\{f_\ell : \ell \in [p]\}, \mathbb{I}} = [0, 1]^p.$$

The non-empty faces in the face lattice $L([0, 1]^p)$ can be enumerated by subsets $\mathcal{U} \subset [p]$ and indices $y_\mathcal{U} \in \times_{\ell \in \mathcal{U}} [2]$ and represented by the cartesian products

$$\mathcal{F}^{(\mathcal{U}, y_\mathcal{U})} = \times_{\ell \in [p]} \mathcal{I}^{l, (\mathcal{U}, y_\mathcal{U})}$$

where

$$\mathcal{I}^{l, (\mathcal{U}, y_\mathcal{U})} = \begin{cases} [0, 1] & \text{if } \ell \notin \mathcal{U} \\ \{y_\ell\} & \text{if } \ell \in \mathcal{U} \end{cases}.$$

Each of these faces can be represented with respect to the elementary graph EL, namely by the tensor product of leg vectors

$$\kappa^{\ell, (\mathcal{U}, y_\mathcal{U})} [Y_\ell] = \begin{cases} \mathbb{I} [Y_\ell] & \text{if } \ell \notin \mathcal{U} \\ \epsilon_{y_\ell} [Y_\ell] & \text{if } \ell \in \mathcal{U} \end{cases}.$$

Example 4 (Simplices) The $(p - 1)$ -dimensional standard simplex $\mathcal{M}_{\Delta, p-1}$ in $[0, 1]^p$ is the convex hull of the vertex sets

$$\mathcal{N} = \{\epsilon_\ell [L] : \ell \in [p]\}.$$

The simplex is the mean polytope of the p -dimensional statistic with

$$t_\ell : [p] \rightarrow [2] \quad , \quad t_\ell [\tilde{\ell}] = \begin{cases} 1 & \text{if } \tilde{\ell} = \ell \\ 0 & \text{if } \tilde{\ell} \neq \ell \end{cases}.$$

The simplex represents all distributions with respect to the trivial base measure $\mathbb{I}$ which coordinates are all 1. When flattened to a vector, any tensor in $\Lambda^\mathbb{I}$ is an element of $\mathcal{M}_{\Delta, p-1}$ where $p = \prod_{k \in [d]} m_k$.

The face lattice of the simplex consists is enumerated by subsets of $[p]$ as

$$L(\mathcal{M}_{\Delta, p-1}) = \{\mathcal{F}^\mathcal{U} : \mathcal{U} \subset [p]\}$$

where the faces are

$$\mathcal{F}^\mathcal{U} = \text{conv}(\epsilon_\ell [L] : \ell \in \mathcal{U})$$

and therefore itself $(|\mathcal{U}| - 1)$ -dimensional standard simplices. The partial order of the faces coincides with the inclusion order of subsets $\mathcal{U}$.

2.5 Partition into Relative Interiors of Faces

Let us now introduce relative interiors, which enables us to find disjoint partitions of the mean polytope.

Definition 5 (Relative Interior) The relative interior $(\mathcal{U})^\circ$ of an arbitrary set $\mathcal{U} \subset \mathbb{R}^p$ is the interior of $\mathcal{U}$ in the affine subspace $\text{conv}(\mathcal{U})$.

Lemma 6 To each $\mu \in \mathcal{M}_{t,\nu}$ there is a unique face $\mathcal{F}$, which we denote by $\mathcal{F}(\mu)$, such that $\mu \in (\mathcal{F})^\circ$.

Proof For any mean vector μ on a face $\mathcal{F}$ we either have $\mu \in (\mathcal{F})^\circ$ or there is a face $\tilde{\mathcal{F}}$ of lower affine dimension with $\tilde{\mathcal{F}} \subset \mathcal{F}$ and $\mu \in \tilde{\mathcal{F}}$. Starting from the face $\mathcal{M}_{t,\nu}$ (for which by assumption $\mu \in \mathcal{M}_{t,\nu}$) we iteratively choose a face $\mathcal{F}$ of lower dimension containing μ until we reach a face with $\mu \in (\mathcal{F})^\circ$. This procedure is guaranteed to terminate, since the affine dimension is decreasing and any face of affine dimension 0 coincides with its relative interior. $\square$

3 Exponential Families

It is well known that exponential families are specific maximum entropy distributions. However, not any maximum entropy distribution is an exponential family member. To prepare for the characterization of generic maximum entropy distributions, we will in this section represent exponential family members as Computation-Activation Networks. We further show that a maximum entropy distribution is a exponential family member, if and only if its mean parameter is in the relative interior of the mean polytope.

3.1 Representation as elementary Computation-Activation Networks

Let us first define exponential families as classes of probability distributions and then derive their representation by elementary Computation-Activation Networks.

Definition 6 (Exponential Family) Given a statistic function

$$t : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}^p$$

and a base measure

$$\nu : \bigtimes_{k \in [d]} [m_k] \rightarrow [0, \infty)$$

with $\langle \nu \rangle_{[\emptyset]} \neq 0$, the set $\Gamma^{t,\nu} = \{\mathbb{P}^{(t,\theta,\nu)} : \theta[L] \in \mathbb{R}^p\} \subset \Lambda^\nu$ of probability distributions

$$\mathbb{P}^{(t,\theta,\nu)}[X_{[d]}] = \frac{\exp \left[\left\langle \sigma^t[X_{[d]}, L], \theta[L] \right\rangle_{[X_{[d]}]} \right]}{\left\langle \exp \left[\left\langle \sigma^t[X_{[d]}, L], \theta[L] \right\rangle_{[X_{[d]}]} \right], \nu[X_{[d]}] \right\rangle_{[\emptyset]}}$$

is called the exponential family to t .

Theorem 2 (Exponential Families are elementary Computation-Activation Networks) Given any base measure ν and a sufficient statistic t we enumerate for each coordinate $\ell \in [p]$ the image $\text{im}(t_\ell)$ by a variable Y_ℓ taking values in $[\text{im}(t_\ell)]$, given an interpretation map

$$I_\ell : [\text{im}(t_\ell)] \rightarrow \text{im}(t_\ell) .$$

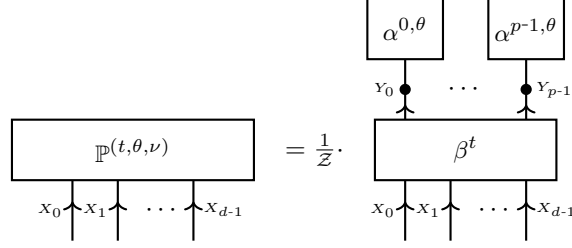


Fig. 2 Representation of a member in an exponential family by a Computation-Activation Network with elementary activation. Here $\mathcal{Z} \in \mathbb{R}$ is a normalization constant.

For any canonical parameter vector $\theta[L] \in \mathbb{R}^p$ we build the activation cores $\alpha^{\ell, \theta}[Y_\ell]$ for each coordinate $y_\ell \in [\text{im}(t_\ell)]$ by

$$\alpha^{\ell, \theta}[Y_\ell = y_\ell] = \exp[\theta[L = \ell] \cdot I_\ell[y_\ell]]$$

and have (see Figure 2)

$$\mathbb{P}^{(t, \theta, \nu)}[X_{[d]}] = \frac{\left\langle \{\beta^t[Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \right\rangle_{[X_{[d]}]}}{\left\langle \{\beta^t[Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \cup \{\nu[X_{[d]}]\} \right\rangle_{[\emptyset]}}.$$

Proof For each $x_{[d]} \in \times_{k \in [d]}[m_k]$ we have

$$\begin{aligned} & \left\langle \{\beta^t[Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \right\rangle_{[X_{[d]}=x_{[d]}]} \\ &= \prod_{\ell \in [p]} \left\langle \beta^{t_\ell}[Y_\ell, X_{[d]}=x_{[d]}], \alpha^{\ell, \theta}[Y_\ell] \right\rangle_{[\emptyset]} \\ &= \prod_{\ell \in [p]} \exp[\theta[L = \ell] \cdot I_\ell[t_\ell[x_{[d]}]]] \\ &= \exp \left[\sum_{\ell \in [p]} \theta[L = \ell] \cdot I_\ell[t_\ell[x_{[d]}]] \right] \\ &= \exp \left[\left\langle \sigma^t[X_{[d]}, L], \theta[L] \right\rangle_{[X_{[d]}=x_{[d]}]} \right]. \end{aligned}$$

Therefore we have

$$\left\langle \{\beta^t[Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : \ell \in [p]\} \right\rangle_{[X_{[d]}]} = \exp \left[\left\langle \sigma^t[X_{[d]}, L], \theta[L] \right\rangle_{[X_{[d]}]} \right].$$

The claim follows, since this implies that also the contraction of both sides with $\nu[X_{[d]}]$ is equivalent. $\square$

We will use the following well known property (see e.g. [32]), that there is a one-to-one map between the canonical parameters and the mean parameters.

Lemma 7 The set of mean parameters of the members of an exponential family is the relative interior of the mean polytope.

Proof See The 3.3 in [33]. $\square$

Based on this property we define the forward and backward mappings to an exponential family.

Definition 7 The forward map of an exponential family is the map

$$F^{(t,\nu)} : \mathbb{R}^p \rightarrow (\mathcal{M}_{t,\nu})^\circ$$

defined as

$$F^{(t,\nu)}(\theta) = \left\langle \mathbb{P}^{t,\theta,\nu}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]}$$

Any map $B^{(t,\nu)} : (\mathcal{M}_{t,\nu})^\circ \rightarrow \mathbb{R}^p$ with $F^{(t,\nu)}(B^{(t,\nu)}(\mu)) = \mu$ for all $\mu \in (\mathcal{M}_{t,\nu})^\circ$ is called a backward map.

3.2 Maximum entropy distributions

We now show that exponential families are maximum entropy distributions.

Lemma 8 Any member of an exponential family $\Gamma^{t,\nu}$ is a maximum entropy distribution with respect to the moment constraint on t and the base measure ν .

Proof Let $\mathbb{P}^{t,\theta,\nu}$ be a exponential family member with canonical parameter θ and let $\mu[L]$ be the corresponding mean parameter. Let $\tilde{\mathbb{P}}[X_{[d]}]$ be any feasible distribution for the maximum entropy problem with respect to the moment constraint on t by $\mu[L]$ and the base measure ν . We also have $\left\langle \tilde{\mathbb{P}}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} = \mu[L]$ and thus

$$\begin{aligned} \mathbb{H}_\nu[\tilde{\mathbb{P}}, \mathbb{P}^{t,\theta,\nu}] &= - \left\langle \tilde{\mathbb{P}}[X_{[d]}], \ln[\mathbb{P}^{t,\theta,\nu}[X_{[d]}]], \nu[X_{[d]}] \right\rangle_{[\emptyset]} \\ &= - \left\langle \tilde{\mathbb{P}}[X_{[d]}], \sigma^t[X_{[d]}, L], \theta[L], \nu[X_{[d]}] \right\rangle_{[\emptyset]} + A^{(t,\nu)}(\theta) \\ &= - \langle \theta[L], \mu[L] \rangle_{[\emptyset]} + A^{(t,\nu)}(\theta) \\ &= \mathbb{H}_\nu[\mathbb{P}^{t,\theta,\nu}]. \end{aligned}$$

With the Gibbs inequality we have if $\tilde{\mathbb{P}} \neq \mathbb{P}^{t,\theta,\nu}$

$$\mathbb{H}_\nu[\mathbb{P}^{t,\theta,\nu}] - \mathbb{H}_\nu[\tilde{\mathbb{P}}] = \mathbb{H}_\nu[\tilde{\mathbb{P}}, \mathbb{P}^{t,\theta,\nu}] - \mathbb{H}_\nu[\tilde{\mathbb{P}}] > 0$$

and thus $\mathbb{H}_\nu[\tilde{\mathbb{P}}] < \mathbb{H}_\nu[\mathbb{P}^{t,\theta,\nu}]$. Therefore, any feasible $\tilde{\mathbb{P}}$ different from $\mathbb{P}^{t,\theta,\nu}$ have less entropy than $\mathbb{P}^{t,\theta,\nu}$ and $\mathbb{P}^{t,\theta,\nu}$ is a maximum entropy distribution. $\square$

Together with Lem. 7 stating the existence of an exponential family member for any mean parameter in the relative interior of the mean polytope we arrive at the following theorem.

Theorem 3 *If and only if $\mu[L]$ is in the relative interior of $\mathcal{M}_{t,\nu}$, then the unique solution of the maximum entropy problem is the distribution*

$$\mathbb{P}^{t,\theta,\nu}[X_{[d]}] \in \Gamma^{t,\nu}$$

where for some backward map $B^{t,\nu}$

$$\theta = B^{t,\nu}(\mu).$$

Proof By Lem. 7 if

$$\mu[L] \in (\mathcal{M}_{t,\nu})^\circ,$$

there is at least one weight parameter $\theta[L] \in \mathbb{R}^p$ with

$$\left\langle \mathbb{P}^{t,\theta,\nu}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} = \mu[L]$$

and $B^{t,\nu}(\mu)$ with respect to any backward map $B^{t,\nu}$ is such one. With Lem. 8 we conclude that $\mathbb{P}^{(t, B^{t,\nu}(\mu), \nu)}$ is the maximum entropy distribution with respect to the moment constraint on t by $\mu[L]$ and the base measure ν . $\square$

Note that while the maximum entropy distribution is unique, the weight vector θ representing the maximum entropy distribution is not always unique. This is reflected by multiple possible choices of backward mappings. Only if the exponential family is minimal (see [33]) then the weight vector is unique. However, we did not restrict to such situations, since exponential families on refined base measures are not minimal.

4 Generic maximum entropy distributions

In the previous section we have shown that the members of an exponential family are maximum entropy distribution. However, their mean parameters build the relative interior of the mean polytope, but are never on faces of smaller dimension. We here characterize the maximum entropy distributions with respect to mean parameters on the full mean polytope.

4.1 Main result

Our main result generalizes the maximum entropy characterization of Thm. 3 to arbitrary mean parameters.

Theorem 4 (Generic characterization of Maximum Entropy Solutions) *Let t be a statistic and ν a base measure. For any $\theta[L]$ and $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ the distribution*

$$\mathbb{P}^{t,\theta,\nu^{t,\mathcal{F}}}[X_{[d]}]$$

is a maximum entropy distribution.

For any $\mu[L]$ the maximum entropy problem has a feasible distribution, if and only if $\mu[L] \in \mathcal{M}_{t,\nu}$. If $\mu[L] \in \mathcal{M}_{t,\nu}$ the maximum entropy distribution is $\mathbb{P}^{t,\theta,\nu^{t,\mathcal{F}}}$ where $\mathcal{F} = \mathcal{F}(\mu)$ (see Lem. 6) and

$$\theta = B^{t,\nu^{t,\mathcal{F}}}(\mu)$$

where $\nu^{t,\mathcal{F}}$ is the refined base measure (see Def. 3) and $B^{t,\nu^{t,\mathcal{F}}}$ is any backward map to the exponential family $\Gamma^{t,\nu^{t,\mathcal{F}}}$ (see Def. 7).

For any hypergraph $\mathcal{G}$ and any face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have that all maximum entropy distributions to mean parameters on $\mathcal{F}^\circ$ are in $\Lambda^{t,\mathcal{G},\nu}$ if $\mathcal{F}$ is representable with respect to $\mathcal{G}$ (see Def. 4).

To prepare for the proof of this theorem we first show in an auxiliary lemma that we can reduce the set of feasible distributions in Problem $\mathbf{P}_{t,\mu,\nu}$.

Lemma 9 For any $\mu[L] \in \mathcal{M}_{t,\nu}$ and a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ with $\mu[L] \in (\mathcal{F})^\circ$ we have that the solutions of $\mathbf{P}_{t,\mu,\nu}$ and $\mathbf{P}_{t,\mu,\nu^{t,\mathcal{F}}}$ coincide.

Proof By Lem. 3 all feasible distributions are representable by the with the face measure refined base measure. We have that any for $\mathbf{P}_{t,\mu,\nu}$ feasible distribution $\mathbb{P}[X_{[d]}]$ satisfies

$$\left\langle \mathbb{P}[X_{[d]}], \nu^{t,\mathcal{F}} \right\rangle_{[\emptyset]} = 1$$

and thus $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$. Conversely, any $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$ satisfies

$$\left\langle \mathbb{P}[X_{[d]}], \nu^{t,\mathcal{F}} \right\rangle_{[\emptyset]} = \left\langle \mathbb{P}[X_{[d]}], \nu \right\rangle_{[\emptyset]} = 1$$

and thus $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu}$. Problem $\mathbf{P}_{t,\mu,\nu}$ is thus equal to

$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}} \mathbb{H}_\nu[\mathbb{P}[X_{[d]}]] \quad \text{subject to} \quad \left\langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} = \mu^*[L]$$

We further have that any $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$

$$\mathbb{H}_\nu[\mathbb{P}[X_{[d]}]] = \mathbb{H}_{\nu^{t,\mathcal{F}}}[\mathbb{P}[X_{[d]}]]$$

and arrive together with the above equivalence at the claim. $\square$

Proof of Thm. 4 To show that for any $\theta[L]$ and $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ the distribution $\mathbb{P}^{t,\theta,\nu^{t,\mathcal{F}}}$ is a maximum entropy distribution, we apply Lem. 9 and Thm. 3.

If $\mu[L] \notin \mathcal{M}_{t,\nu}$ then Problem $\mathbf{P}_{t,\mu,\nu}$ is not feasible by definition. In case $\mu[L] \in \mathcal{M}_{t,\nu}$ we find $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ with $\mathcal{F} = \mathcal{F}(\mu)$ and by Lem. 9 the solution of maximum entropy problem Problem $\mathbf{P}_{t,\mu,\nu}$ coincides with $\mathbf{P}_{t,\mu,\nu^{t,\mathcal{F}}}$. Since by Lem. 2 the face is the polytope

$$\mathcal{F} = \mathcal{M}_{t,\nu^{t,\mathcal{F}}}$$

we find a weight parameter

$$\theta = B^{t, \nu^t, \mathcal{F}}(\mu).$$

and by Thm. 3 $\mathbb{P}^{t, \theta, \nu^t, \mathcal{F}}$ is the corresponding maximum entropy distribution.

To show the last claim, it is enough to show that for any by $\mathcal{G}$ representable face $\mathcal{F}$ and for any $\theta \in \mathbb{R}^p$ we have

$$\mathbb{P}^{t, \theta, \nu^t, \mathcal{F}} \in \Lambda^{t, \mathcal{G}, \nu}.$$

To this end, we construct activation cores as follows. By Def. 4 and Lem. 4 there is a tensor network $\kappa^{\mathcal{G}}$ such that

$$\nu^{t, \mathcal{F}}[X_{[d]}] = \langle \kappa^{\mathcal{G}} \cup \{\nu[X_{[d]}]\} \rangle_{[X_{[d]}]}.$$

For each $\ell \in [p]$ we find an edge $e(\ell) \in \mathcal{E}$ with $\ell \in e(\ell)$. Now we define a tensor network $\tau^{\mathcal{G}}$ by the cores to $e \in \mathcal{E}$ as

$$\tau^e[X_e] = \langle \{\kappa^e[X_e]\} \cup \{\alpha^{\ell, \theta}[Y_{\ell}] : e = e(\ell)\} \rangle_{[X_e]}.$$

Then, we have

$$\left\langle \beta^t[Y_{[p]}, X_{[d]}], \langle \tau^{\mathcal{G}} \rangle_{[Y_{[p]}]} \right\rangle_{[X_{[d]}]} = \left\langle \beta^t[Y_{[p]}, X_{[d]}], \alpha^{\theta}[Y_{[p]}], \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]} \right\rangle_{[X_{[d]}]}$$

and therefore

$$\mathbb{P}^{t, \theta, \nu^t, \mathcal{F}} = \frac{\langle \{\beta^t[Y_{[p]}, X_{[d]}]\} \cup \tau^{\mathcal{G}} \rangle_{[X_{[d]}]}}{\langle \{\beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}]\} \cup \tau^{\mathcal{G}} \rangle_{[\emptyset]}}.$$

We conclude $\mathbb{P}^{t, \theta, \nu^t, \mathcal{F}} \in \Lambda^{t, \mathcal{G}, \nu}$.

□

4.2 Family of maximum entropy distributions

The family of maximum entropy distributions is the set

$$\Gamma^{t, \nu} = \{\mathbb{P}[X_{[d]}] : \exists \mu \in \mathcal{M}_{t, \nu} : \mathbb{P}[X_{[d]}] \text{ is a solution of Problem } \mathbf{P}_{t, \mu, \nu}\}.$$

Thm. 4 suggest a parametrization of this family as tuples of corresponding faces $\mathcal{F}$ and weight vectors $\mu[L]$. As sketched in Figure 3 the activation tensor of each maximum entropy distribution is decomposed into a possibly non-elementary boolean tensor corresponding to the base measure refinement on a face $\mathcal{F}$, and an elementary tensor dependent on the weight vector $\theta \in \mathbb{R}^p$. We thus define the canonical parameter space

$$\mathcal{P}_{t, \nu} := L(\mathcal{M}_{t, \nu}) \times \mathbb{R}^p$$

and have

$$\Gamma^{t, \nu} = \left\{ \frac{\langle \alpha^{\theta}[Y_{[p]}], \kappa^{\mathcal{F}}[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}] \rangle_{[X_{[d]}]}}{\langle \alpha^{\theta}[Y_{[p]}], \kappa^{\mathcal{F}}[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}] \rangle_{[\emptyset]}} : \theta[L] \in \mathbb{R}^p, \mathcal{F} \in \mathcal{P}_{t, \nu} \right\}.$$

Note that these families are disjoint, since each member of an exponential family has support by the support of the face measure.

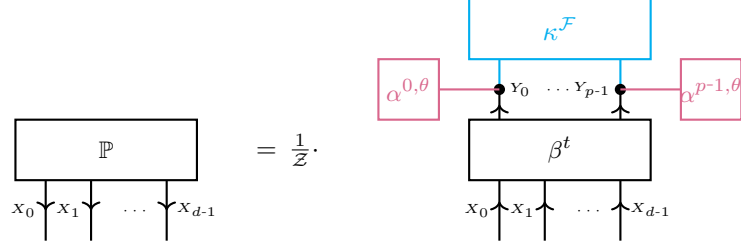


Fig. 3 Tensor network decomposition of maximum entropy distributions to the constraint $\mu[L] = \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L] \rangle_{[L]}$. Blue: Boolean tensor $\kappa^{\mathcal{F}}[Y_{[p]}]$ dependent on the face $\mathcal{F}(\mu)$, which represents the refinement of the base measure to the face, and is possibly non-elementary. Red: Elementary activation tensor $\alpha^\theta[Y_{[p]}] = \bigotimes_{s \in [n]} \alpha^{\ell, \theta}[Y_\ell]$ to the weight $\theta[L] = B^{t, \nu^t, \mathcal{F}}(\mu)$.

5 Boolean statistics

We say that a statistic is boolean, if $n_\ell = 2$ for all $\ell \in [p]$. The mean polytope is then a 0/1-polytope [34], which vertices are exactly the contained boolean vectors. The special case of boolean statistics is especially interesting for two reasons. First, we can provide additional intuition on those faces representable by elementary face activating tensors, namely based on intersections with cube faces. Based on these properties we also show that any elementary Computation-Activation Network to a boolean statistic is a maximum entropy distribution. Second, the statistic itself can be interpreted in terms of logical formulas. We exploit this property to provide examples and to characterize each 0/1-polytope as a mean polytope to a collection of propositional formulas.

5.1 Characterization of elementary representable faces

Orienting on Example 3 we define cube-likeness of faces and polytopes.

Definition 8 We say that a face $\mathcal{F}$ of $\mathcal{M}_{t, \nu}$ is cube-like, if it is empty or there is $\mathcal{U} \subset [p]$ and $y_{\mathcal{U}} \in \times_{\ell \in \mathcal{U}} [2]$ such that

$$\mathcal{F} = \mathcal{M}_{t, \nu} \cap \mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}.$$

Here we denote by $\mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$ the faces of the hypercube $[0, 1]^p$ (see Example 3). We further say that a polytope $\mathcal{M}_{t, \nu}$ is cube-like, if all faces $\mathcal{F} \in L(\mathcal{M}_{t, \nu})$ are cube-like.

We now show that a face is cube-like if and only if it is representable by an elementary tensor.

Theorem 5 Let t be a boolean statistic, ν a base measure and $\mathcal{F}$ be a face of $\mathcal{M}_{t, \nu}$. Then the following are equivalent:

- (i) $\mathcal{F}$ is cube-like (see Def. 8).
- (ii) $\mathcal{F}$ is representable by an elementary tensor (see Def. 4).

Proof If the face is empty, i.e. $\mathcal{F} = \emptyset$, it is by definition cube-like and has $0 \left[Y_{[p]} \right]$ as an elementary activation tensor. We therefore assume in the following $\mathcal{F} \neq \emptyset$.

(i) $\Rightarrow$ (ii): Let us assume that $\mathcal{F}$ is cube-like, that is there is $\mathcal{U} \subset [p]$ and $y_{\mathcal{U}} \in \times_{\ell \in \mathcal{U}} [2]$ such that $\mathcal{F} = \mathcal{M}_{t,\nu} \cap \mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$. We use that $\mathcal{F}$, $\mathcal{M}_{t,\nu}$ and $\mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$ are the convex hulls of the cube vertex sets $\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}}$, $\mathcal{N}_{t,\mathcal{M}}$ and

$$\mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}} = \{v[L] : \forall \ell \in \mathcal{U} : v[L = \ell] = y_{\mathcal{U}}\}, \quad (1)$$

which implies that

$$\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}} = \mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}} \cap \mathcal{N}_{t,\mathcal{M}}.$$

Thus, we can choose $\mathcal{U} = \mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}}$ for the representation of the face $\mathcal{F}$ (see Def. 4) and further have that

$$\sum_{v[L] \in \mathcal{N}^{(\mathcal{U}, y_{\mathcal{U}})}} \epsilon_{v[L]} \left[Y_{[p]} \right] = \bigotimes_{\ell \in [p]} \kappa^{\ell, (\mathcal{U}, y_{\mathcal{U}})} [Y_{\ell}].$$

We have thus found an elementary activation tensor for the face $\mathcal{F}$.

(ii) $\rightarrow$ (i) Conversely, let $\xi \left[Y_{[p]} \right]$ be an elementary activation tensor of the face $\mathcal{F}$ in $\mathcal{M}_{t,\nu}$. Since $\xi \left[Y_{[p]} \right]$ is the sum of different one-hot encodings it is boolean and we find an elementary decomposition $\xi \left[Y_{[p]} \right] = \bigotimes_{\ell \in [p]} \xi^{\ell} [Y_{\ell}]$ such that the leg vectors $\xi^{\ell} [Y_{\ell}]$ are boolean. Since $\mathcal{F} \neq \emptyset$ we further have for $\ell \in [p]$ that $\xi^{\ell} [Y_{\ell}] \neq 0 [Y_{\ell}]$, and thus $\xi^{\ell} [Y_{\ell}] \in \{\epsilon_0 [Y_{\ell}], \epsilon_1 [Y_{\ell}], \mathbb{I} [Y_{\ell}]\}$. We construct a set $\mathcal{U} \subset [p]$

$$\mathcal{U} = \left\{ \ell : \xi^{\ell} [Y_{\ell}] \neq \mathbb{I} [Y_{\ell}] \right\}$$

and a tuple

$$y_{\ell} = \begin{cases} 1 & \text{if } \xi^{\ell} [Y_{\ell}] = \epsilon_1 [Y_{\ell}] \\ 0 & \text{if } \xi^{\ell} [Y_{\ell}] = \epsilon_0 [Y_{\ell}] \end{cases}.$$

By construction $\xi^{\ell} [Y_{\ell}] = \kappa^{\ell, (\mathcal{U}, y_{\mathcal{U}})} [Y_{\ell}]$ (see Example 3) follows for all $\ell \in [p]$ and therefore $\xi \left[Y_{[p]} \right] = \kappa^{(\mathcal{U}, y_{\mathcal{U}})} \left[Y_{[p]} \right]$. We therefore have for the set (1)

$$\xi \left[Y_{[p]} \right] = \sum_{v \in \mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}}} \epsilon_v \left[Y_{[p]} \right]$$

and since $\xi \left[Y_{[p]} \right]$ is an activation tensor for $\mathcal{F}$, that

$$\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}} = \mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}} \cap \mathcal{N}_{t,\mathcal{M}}.$$

Taking convex hulls on both sides, this is equivalent to

$$\mathcal{F} = \mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})} \cap \mathcal{M}_{t,\nu}.$$

We conclude that $\mathcal{F}$ is cube-like. $\square$

Thm. 5 implies that the maximum entropy distributions to hypercubes (see Example 3) and simplices (see Example 4) are in $\Lambda^{t, \text{EL}, \nu}$, since these polytopes are cube-like.

We can further show, that for boolean statistics, any distribution in $\Lambda^{t, \text{EL}, \nu}$ is a maximum entropy distribution.

Theorem 6 *Let t be a boolean statistic. Any distribution in $\Lambda^{t, \text{EL}, \nu}$ is a maximum entropy distribution with respect to (t, μ, ν) where μ is its mean parameter.*

Proof Let $\mathbb{P} \in \Lambda^{t, \text{EL}, \nu}$ be arbitrary. We then find an elementary tensor $\bigotimes_{s \in [n]} \tau^\ell[Y_\ell]$ and $\mathcal{Z} \in \mathbb{R}$ such that

$$\mathbb{P}[X_{[d]}] = \frac{1}{\mathcal{Z}} \left\langle \beta^t[Y_{[p]}, X_{[d]}], \bigotimes_{s \in [n]} \tau^\ell[Y_\ell] \right\rangle_{[X_{[d]}]}.$$

We then define for each $\ell \in [p]$ the vector $\kappa^\ell[Y_\ell]$ as the support of $\tau^\ell[Y_\ell]$ and

$$\theta[L = \ell] = \begin{cases} \ln \left[\frac{\tau^\ell[Y_\ell=1]}{\tau^\ell[Y_\ell=0]} \right] & \text{if } \tau^\ell[Y_\ell=0], \tau^\ell[Y_\ell=1] \neq 0 \\ 0 & \text{else} \end{cases}.$$

Since t is a boolean statistic the $\bigotimes_{\ell \in [p]} \kappa^\ell[Y_\ell]$ is a face activation tensor to the face

$$\mathcal{F} := \mathcal{M}_{t, \mathbb{I}} \cap \mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$$

where $\mathcal{U} = \{\ell : \kappa^\ell[Y_\ell] \neq \mathbb{I}[Y_\ell]\}$ and for $\ell \in \mathcal{U}$

$$y_\ell = \begin{cases} 0 & \text{if } \kappa^\ell[Y_\ell=1] = 0 \\ 1 & \text{else} \end{cases}.$$

Therefore, $\mathbb{P}$ is the maximum entropy distribution to the canonical parameters $\mathcal{F}$ and θ . $\square$

5.2 Interpretation by propositional formulas

We can understand each feature as a propositional formula and the variables $X_{[d]}$ as atoms (possibly after a binarization). We use propositional connectives for syntactical representations of the formulas and denote by $\neg^1$ the logical negation and by $\neg^0$ the identity. A boolean tensor $f[X_{[d]}]$ is called satisfiable, if there is at least one $x_{[d]} \in \times_{k \in [d]} [m_k]$ with $f[X_{[d]} = x_{[d]}] = 1$.

Theorem 7 *Let t be a boolean statistic. Any boolean vector $v[L] \in \{0, 1\}^p$ is a vertex of $\mathcal{M}_{t, \mathbb{I}}$ if and only if*

$$f_v[X_{[d]}] := \bigwedge_{\ell \in [p]} \neg^{1-v[L=\ell]} f_\ell[X_{[d]}]$$

is satisfiable.

Proof Let $v[L] \in \{0, 1\}^p$ be arbitrary. If and only if $v[L]$ is a vertex of $\mathcal{M}_{t, \mathbb{I}}$ we find $x_{[d]} \in \times_{k \in [d]} [m_k]$ with $v[L] = \sigma^t X_{[d]} = x_{[d]}, L$. This is further equivalent to $f_\ell[X_{[d]} = x_{[d]}] = v[L = \ell]$ for any $\ell \in [p]$, $\neg^{1-v[L=\ell]} f_\ell[X_{[d]} = x_{[d]}] = 1$ for any $\ell \in [p]$, and $f_v[X_{[d]} = x_{[d]}] = 1$. $\square$

Since the presence of boolean vectors in $\mathcal{M}_{t,\mathbb{I}}$ is by Thm. 7 related to the satisfiability of the conjunction $f_v[X_{[d]}]$, the absence is related to entailment statements. To be more precise, for any $v[L] \in \{0, 1\}^p$ and arbitrary $\mathcal{U} \subset [p]$ we have $v[L] \notin \mathcal{M}_{t,\mathbb{I}}$, if and only if

$$\left(\bigwedge_{\ell \in \mathcal{U}} \neg^{1-v[L=\ell]} f_\ell[X_{[d]}] \right) \models \left(\bigvee_{\ell \notin \mathcal{U}} \neg^{v[L=\ell]} f_\ell[X_{[d]}] \right).$$

By definition (see [35]) the latter holds if and only if

$$\left(\bigwedge_{\ell \in \mathcal{U}} \neg^{1-v[L=\ell]} f_\ell[X_{[d]}] \right) \wedge \neg \left(\bigvee_{\ell \notin \mathcal{U}} \neg^{v[L=\ell]} f_\ell[X_{[d]}] \right) = f_v[X_{[d]}]$$

is not satisfiable.

Atomic formulas and minterm formulas correspond to hypercubes and simplices as we show next.

Example 5 (Atomic formulas) Let us consider the case of atomic formulas $f_\ell[X_{[d]}]$ defined with the coordinates at $x_{[d]} \in \times_{k \in [d]} [m_k]$ by

$$f_\ell[X_{[d]} = x_{[d]}] = x_\ell.$$

The mean polytope in this case is the d -dimensional hypercube (see Example 3)

$$\mathcal{M}_{\mathcal{F}_{[d]}, \mathbb{I}} = [0, 1]^p$$

which is called a simple polytope, since each vertex is contained in the minimal number of d facets.

Each face is characterized by the projections onto each variable, which is either $\{0\}$, $\{1\}$ or $[0, 1]$. The projections are represented by the tuple $(\mathcal{U}, y_\mathcal{U})$ defined in the following way:

- We define the set $\mathcal{U} \subset [d]$ of variables, such that the projection onto the variable is $\{0\}$ or $\{1\}$
- We define to each $\ell \in \mathcal{U}$ an index $y_\ell = 0$ if the projection is $\{0\}$ and $y_\ell = 1$ if the projection is $\{1\}$.

Trivially, each face of the hypercube is a cube face.

More general, the mean polytope to a collection $f_\ell[X_{[d]}]$ of propositional formulas is the hypercube, if and only if for any boolean vector $v[L]$ we have

$$\left\langle \bigwedge_{\ell \in [p]} \neg^{1-v[L=\ell]} f_\ell[X_{[d]}] \right\rangle_{[\emptyset]} \neq 0.$$

This means that no conjunction of possibly negated formulas from this statistic entails nor contradicts another formula.

Example 6 (Minterm formulas build simplices) The set of minterm formulas is indexed by $v[L] \in \{0, 1\}^p$ and given by

$$f_v[X_{[d]}] = \bigwedge_{\ell \in [d]} \neg^{1-v[X_{[d]}=x_{[d]}]} X_\ell = \epsilon_{v[L]}[X_{[d]}]$$

where by X_ℓ we denote the ℓ -th atomic formula (see Example 5). The mean polytope to the minterm statistic and base measure ν is the standard simplex of dimension

$$\left| \left\{ v[L] : v \in \{0, 1\}^p, \nu[X_{[d]} = v[L]] \neq 0 \right\} \right| - 1,$$

that is the set

$$\mathcal{M}_{\mathcal{F}_\wedge, \nu} = \text{conv} \left(\epsilon_{v[L]}[Y_{[p]}] : v \in \{0, 1\}^p, \nu[X_{[d]} = v[L]] \neq 0 \right).$$

We notice that for boolean base measures $\mathcal{M}_{\mathcal{F}_\wedge, \nu} = \Delta^\nu$. The simplex is further cube-like, since each face is indexed by a subset $\mathcal{U} \subset \{0, 1\}^p$ of vertices and the intersection of the simplex with the cube face $\mathcal{F}^{\{0, 1\}^p \setminus \mathcal{U}, 1_{\{0, 1\}^p \setminus \mathcal{U}}}$.

More general the mean polytope to a boolean statistic is a standard simplex, if and only if each target coordinate restriction $f_{\tilde{\ell}}$ contradicts all others, that is

$$\forall \ell \neq \tilde{\ell} \in [p] : f_\ell \models \neg f_{\tilde{\ell}}.$$

Let us now provide examples of statistics, which polytopes are not cube-like.

Example 7 (Maximum entropy distribution with non-elementary activation cores) Consider two atomic variables X_0 and X_1 and a statistic $\mathcal{F}$ consisting in the formulas

$$f_0[X_0, X_1] = (X_0 \wedge X_1) \quad , \quad f_1[X_0, X_1] = (X_0 \Rightarrow X_1)$$

with the coordinatewise expressions

$$f_0[X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] \quad , \quad f_1[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1].$$

We can think of X_0 as a feature on an invoice, and X_1 as a feature on the accounting proposal.

From this we have

$$\begin{aligned} \beta^{(f_0, f_1)}[Y_0 = 0, Y_1 = 0, X_0, X_1] &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} [X_0, X_1] \quad , \quad \beta^{(f_0, f_1)}[Y_0 = 0, Y_1 = 1, X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} [X_0, X_1] \quad , \\ \beta^{(f_0, f_1)}[Y_0 = 1, Y_1 = 0, X_0, X_1] &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} [X_0, X_1] \quad \text{and} \quad \beta^{(f_0, f_1)}[Y_0 = 1, Y_1 = 1, X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1]. \end{aligned}$$

Since the only vanishing slice of $\beta^{\mathcal{F}}$ with respect to the head variables is that to $y_{0,1} = (1, 0)$, the vertices of the mean polytope are the vectors to the other head indices. The mean polytope is the convex hull of these vertices

$$\mathcal{M}_{(f_0, f_1)} = \text{conv} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} [L], \begin{pmatrix} 0 \\ 1 \end{pmatrix} [L], \begin{pmatrix} 1 \\ 1 \end{pmatrix} [L] \right).$$

This polytope has a non cube-like face (sketched blue in Figure 4), which is the convex hull of the vertices $(00)^T, (11)^T$. This face is parametrized by the (CP-rank 2) hard activation core

$$\kappa^{(0,0),(1,1)}[Y_0, Y_1] = \epsilon_{(0,0)}[Y_0, Y_1] + \epsilon_{(1,1)}[Y_0, Y_1] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} [Y_0, Y_1]$$

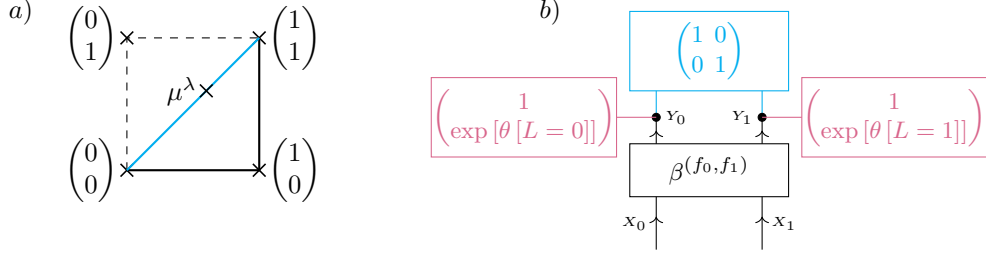


Fig. 4 a) Mean polytope of the statistic $t = (X_0 \wedge X_1, X_0 \Rightarrow X_1)$ (thick), as a subset of the cube $[0, 1]^2$ (dashed). The blue line is the face of the polytope, which is not cube like, that is not an intersection of the polytope with the faces of the polytope. We further define for $\lambda \in (0, 1)$ a mean parameter $\mu_\lambda [L] = [\lambda \ \lambda]^T$ which is on the interior of the blue face. b) Corresponding Computation-Activation Network being the maximum entropy distribution reproducing $\mu_\lambda [L]$, when λ is the sigmoid of $\theta [L = 0] + \theta [L = 1]$.

and has the refined base measure

$$\left\langle \kappa^{(0,0),(1,1)} [Y_0, Y_1], \beta^{\mathcal{F}} [Y_0, Y_1, X_0, X_1] \right\rangle_{[X_0, X_1]} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} [X_0, X_1] .$$

Any mean parameter μ on the interior of that face can be parametrized by a scalar $\lambda \in (0, 1)$

$$\mu_\lambda [L] = (\lambda \ \lambda) [L] .$$

With the canonical parameters $\theta [L] \in \mathbb{R}^2$ of the maximum entropy distributions on this face by

$$\mathbb{P} [X_0, X_1] = \frac{1}{1 + \exp [\theta [L = 0] + \theta [L = 1]]} \begin{pmatrix} 0 & 0 \\ 1 & \exp [\theta [L = 0] + \theta [L = 1]] \end{pmatrix} [X_0, X_1]$$

we get the correspondence by the sigmoid

$$\lambda = \frac{1}{1 + \exp [-(\theta [L = 0] + \theta [L = 1])]} .$$

Note, that the hard activation core $\kappa^{(0,0),(1,1)} [Y_0, Y_1]$ to the blue face is the only non-elementary activation core. While the vertices have always elementary cores, the further non-vertex faces have elementary activation cores

$$\kappa^{(0,0),(1,0),(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} [Y_0, Y_1] = \mathbb{I} [Y_0] \otimes \mathbb{I} [Y_1] , \quad \kappa^{(0,0),(1,0)} [Y_0, Y_1] = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} [Y_0, Y_1] = \mathbb{I} [Y_0] \otimes \epsilon_0 [Y_1] ,$$

$$\kappa^{(1,0),(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} [Y_0, Y_1] = \epsilon_0 [Y_0] \otimes \mathbb{I} [Y_1] .$$

The maximum entropy distributions to mean parameters on the interior of all other faces than the blue face are represented by Computation-Activation Networks with only elementary activation cores. The corresponding lattice is sketched in Figure 5.

5.3 Faces activated in the TT format

We further present examples of faces in a TT format representation. This format describes tensor networks on a hypergraph (see Figure 6)

$$\text{TT} = (X_{[d]} \cup I_{[d-1]}, \{ \{ I_{k-1}, X_k, I_k \} : k \in \{1, \dots, d-2\} \} \cup \{ \{ X_0, I_0 \}, \{ I_{d-2}, X_{d-1} \} \}) .$$

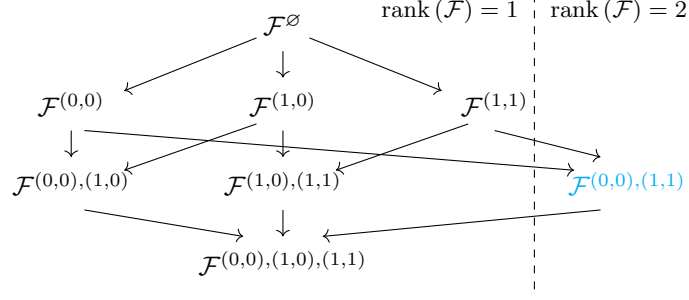


Fig. 5 Face lattice $L((X_0 \wedge X_1, X_0 \Rightarrow X_1))$ to Example 7. The directed arrows represent inclusion of the faces, which is a partial order of the faces. The face $\mathcal{F}^{(0,0),(1,1)}$ is the only face, which is not representable by an elementary hypergraph. It is representable in a CP with hidden rank 2

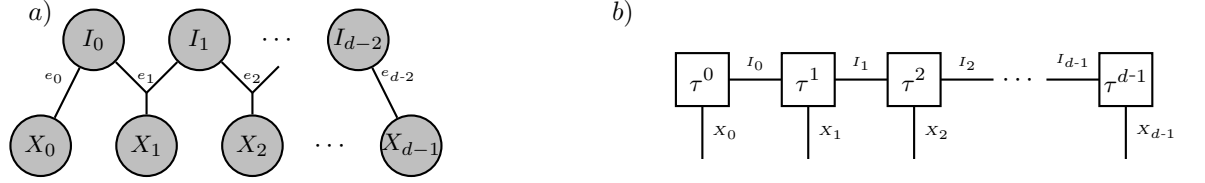


Fig. 6 TT format consisting in the TT hypergraph a) and decorating tensor networks b).

Example 8 (TT representation of diagonal faces) Consider the vertex set

$$\mathcal{N} := \{0, 1\}^p / v[L]$$

for some boolean vector $v[L]$. We are interested in the face $\mathcal{F}^{\triangleleft, 1}$ with the normal $\theta[L] = \frac{1}{2}(2v[L] - \mathbb{I}[L])$. Its vertices are the p elements of $\{0, 1\}^p$, which differ from $v[L]$ in exactly one coordinate (i.e. those with Hamming distance of 1 from $v[L]$). We denote these vertices by v^ℓ for $\ell \in [p]$, which coordinates to $\tilde{\ell} \in [p]$ are

$$v^\ell[L = \tilde{\ell}] = \begin{cases} v[L = \tilde{\ell}] & \text{if } \tilde{\ell} \neq \ell \\ 1 - v[L = \tilde{\ell}] & \text{if } \tilde{\ell} = \ell \end{cases}.$$

We now represent their sum in an TT with hidden ranks $r_0 = \dots = r_{p-2} = 2$. The boolean hidden variables are denoted by $I_{[p-1]}$ and are interpreted as indicators, whether the coordinate flip has happened in $[\ell]$ coordinates. We now construct a TT core for $\ell = 0$ by

$$\tau^0[Y_0, I_0] = \epsilon_{v[L=0]}[Y_0] \otimes \epsilon_0[I_0] + \epsilon_{1-v[L=0]}[Y_0] \otimes \epsilon_1[I_0]$$

and for $\ell \notin \{0, p-1\}$ the cores

$$\begin{aligned} \tau^\ell[I_{\ell-1}, Y_\ell, I_\ell] &= \epsilon_0[I_{\ell-1}] \otimes \epsilon_{1-v[L=\ell]}[Y_\ell] \otimes \epsilon_1[I_\ell] + \epsilon_0[I_{\ell-1}] \otimes \epsilon_{v[L=\ell]}[Y_\ell] \otimes \epsilon_0[I_\ell] \\ &\quad + \epsilon_1[I_{\ell-1}] \otimes \epsilon_{v[L=\ell]}[Y_\ell] \otimes \epsilon_1[I_\ell] \end{aligned}$$

and for $\ell = p-1$

$$\tau^{p-1}[I_{p-1}, Y_{p-1}] = \epsilon_0[I_{p-1}] \otimes \epsilon_{1-v[L=p-1]}[Y_{p-1}] + \epsilon_1[I_{p-1}] \otimes \epsilon_{v[L=p-1]}[Y_{p-1}].$$

For this tensor network in the TT format we have

$$\sum_{\ell \in [p]} \epsilon_{v^\ell} [Y_{[p]}] = \left\langle \{\tau^0 [Y_0, I_0], \tau^{p-1} [I_{p-2}, Y_{p-1}]\} \cup \{\tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] : \ell \in \{1, \dots, p-2\}\} \right\rangle_{[Y_{[p]}]}.$$

The TT multirank of 2 is furthermore minimal, since each matrifcation based on a partition of $[p]$ into non-empty sets has a matrix rank of 2.

5.4 Construction of boolean statistics to polytopes

We now show, that any convex polytope with boolean vertices in $\mathbb{R}^p$ is the mean polytope to a boolean statistic. To construct the corresponding formulas we denote by $\mathcal{M}|_{\mathcal{U}}$ the restriction of the vectors in the set $\mathcal{M}$ to the coordinates in $\mathcal{U}$.

Theorem 8 *Let $\mathcal{M}$ an arbitrary polytope with boolean vertices in $\mathbb{R}^p$. Then we construct propositional formulas on atoms $X_{[p]}$ by*

$$f_0 [X_0] = \begin{cases} \top & \text{if } \mathcal{M}|_{\{0\}} = \{1\} \\ \perp & \text{if } \mathcal{M}|_{\{0\}} = \{0\} \\ X_0 & \text{if } \mathcal{M}|_{\{0\}} = [0, 1] \end{cases}$$

and iteratively for $\ell \in [p]$ with $\ell \geq 1$ by

$$f_\ell [X_{[\ell+1]}] = \bigwedge_{v[L] \in (\mathcal{M} \cap \{0,1\}^p)|_{[\ell]}} \left(\left(\bigwedge_{\tilde{\ell} \in [\ell]} \neg^{1-v[L=\tilde{\ell}]} f_{\tilde{\ell}} [X_{[\tilde{\ell}+1]}] \right) \Rightarrow \begin{cases} \top & \text{if } (\mathcal{M} \cap v \times \mathbb{R}^{p-\ell})|_{\{\ell\}} = \{1\} \\ \perp & \text{if } (\mathcal{M} \cap v \times \mathbb{R}^{p-\ell})|_{\{\ell\}} = \{0\} \\ X_\ell & \text{if } (\mathcal{M} \cap v \times \mathbb{R}^{p-\ell})|_{\{\ell\}} = [0, 1] \end{cases} \right).$$

Here we denote by $\mathcal{M}|_V$ the projections of the vertices in $\mathcal{M}$ onto the subspaces V , and by 0_p the zero vector in $\mathbb{R}^p$.

Proof We show per induction, that for any $\ell \in [p]$ the family of Hybrid Logic Networks with the statistic $f_{[\ell+1]}$ by the first $\ell + 1$ formulas has the mean polytope

$$\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}} = \mathcal{M}|_{[\ell]}. \quad (2)$$

" $\ell = 0$ ": The polytope $\mathcal{M}|_{[\ell]} = \{1\}$ (respectively $\mathcal{M}|_{[\ell]} = \{0\}$) is reproduced by f_0 being a tautology (respectively a contradiction). In the case $\mathcal{M}|_{[\ell]} = [0, 1]$ the polytope is reproduced by the any formula, which is neither a tautology nor a contradiction, and the atomic formula X_0 is an example of such an contingency. Since the projection of a 0-1 polytope onto the first coordinates is itself a 0-1 polytope, these are the only possible cases and we conclude that in all

$$\mathcal{M}_{f_{[\ell]}, \mathbb{I}} = \mathcal{M}|_{[\ell]}.$$

" $\ell \rightarrow \ell + 1$ ": We define for any $v \in \mathcal{M}_{f_{[\ell]}, \mathbb{I}} \cap \{0, 1\}^\ell$ the formula

$$f_{\ell, v} [X_{[\ell+1]}] := \bigwedge_{\tilde{\ell} \in [\ell]} \neg^{1-v[L=\tilde{\ell}]} f_{\tilde{\ell}} [X_{[\tilde{\ell}+1]}].$$

and notice that all formulas are satisfiable (since the corresponding vertices are in the $\mathcal{M}_{f_{[\ell]}, \mathbb{I}}$) and have disjoint models. We have for any boolean vector $v \in \{0, 1\}^\ell$ and $\lambda \in \{0, 1\}$ that $v \times \{\lambda\} \in \mathcal{M}|_{[\ell+1]}$, if and only if

$$v \in \mathcal{M}|_{[\ell]} \quad \text{and} \quad \lambda \in \mathcal{M} \cup v \times \mathbb{R}^{p-\ell}|_{[\ell]}.$$

With modus ponens on $f_{\ell+1} [X_{[\ell+2]}]$ this is further equivalent to the satisfaction of

$$f_{\ell,v} [X_{[\ell+1]}] \wedge \neg^{1-\lambda} f_{\ell+1} [X_{[\ell+2]}]$$

and thus to $v \times \{\lambda\} \in \mathcal{M}_{f_{[\ell+1]}, \mathbb{I}}$. The vertices of the 0/1 polytopes $\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}}$ and $\mathcal{M}_{[\ell+1]}$ thus coincide and both polytopes are therefore equivalent.

By induction the equation (2) holds for arbitrary $\ell \in [p]$. For $\ell = p-1$ this is the claim. $\square$

5.5 Examples in statistical physics and coding theory

Let us now investigate cube-like faces to standard examples in statistical physics and coding theory.

Example 9 (Ising model on 2 nodes) Consider two boolean variables X_0, X_1 and the Ising statistic $t = (f_0, f_1, f_2)$ by three propositional formulas

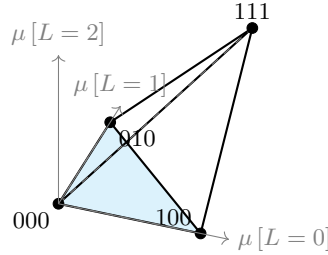
$$f_0 [X_{[2]}] = X_0 \quad , \quad f_1 [X_{[2]}] = X_1 \quad \text{and} \quad f_2 [X_{[2]}] = X_0 \wedge X_1 .$$

In the Ising interpretation, the boolean variables represent interacting spins at two locations. Their value is then measured by the first two formulas and their interaction by the third.

The vertices of the mean polytope to this statistic are

$$\begin{aligned} \sigma^t [X_{[2]} = (0, 0), L] &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} [L] \quad , \quad \sigma^t [X_{[2]} = (0, 1), L] = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} [L] \quad , \\ \sigma^t [X_{[2]} = (1, 0), L] &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} [L] \quad \text{and} \quad \sigma^t [X_{[2]} = (1, 1), L] = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} [L] \quad . \end{aligned}$$

The mean polytope is the convex hull of these, sketched as:



We notice, that the convex hull of any three out of the four vertices builds a facet. The only cube-like facet of these four (sketched in cyan in the above plot) is the convex hull of

$$\text{conv} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} [L], \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} [L], \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} [L] \right)$$

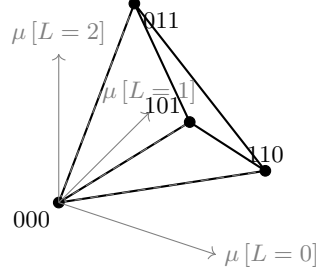
to which we have the parametrization by $(\mathcal{U}, y_{\mathcal{U}}) = (\{2\}, 0)$.

As a consequence, there are maximum entropy distributions of mean parameters in the Ising statistics, which do not have a representation by an elementary CompActNet in the Ising statistic.

Example 10 (Codeword Polytopes) Consider the codeword polytope with the vertices

$$\mathcal{N} = \left\{ y_{[3]} : \sum_{k \in [3]} y_k \mod 2 = 0 \right\} = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} [L], \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} [L], \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} [L], \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} [L] \right\}$$

We interpret the vertices by the strings of length 3, which parity vanishes. Such constrained are common in the construction of codewords using parity-check codes (see [36]).



All facets are non-cube-like, whereas all other faces are cube-like (the edges are labeled by the 6 facets of $[0, 1]^3$).

Each facet is characterized by the Hamming distance of exactly 1 to a single of the 4 cube vertices, which are not codewords (see Example 3.9 in [33]). We can construct TT activation tensors with rank 2, which are calculating in the hidden ranks the Hamming distance to these vertex. Each core tensor in that format adds the contribution of a mode to the Hamming distance.

In more generality, Low-Density-Parity-Checking Codes (see [36]) construct codewords by a collection of parity constraints on subsets of variables. Such polytopes are also known as cycle polytopes, and studied e.g. in [37].

6 Outlook

This work characterized all discrete maximum entropy distributions as Computation-Activation Networks, a class of tensor networks. The mean polytope contains those mean parameters, to which entropy maximization problems are feasible. While exponential families are known to be the maximum entropy distributions to the interior of the polytope, we showed that refining the base measure with respect to faces enables the representation of maximum entropy distributions to mean parameters on the particular face. To this end we utilized the tensor network formalism and showed how to represent exponential families and their generalizations to maximum entropy distributions.

One important application is estimation of maximum entropy distributions from data, such as learning in exponential families and graphical models [38]. Maximum likelihood problems are known to be dual to maximum entropy problems [2] and therefore face similar representation challenges. Here the mean parameters of empirical distributions serve as sufficient statistics and are matched by trainable weights in algorithms such as iterative proportional fitting [39, 40]. Optimizing parameters within a poor choice of learning architecture will lead to numerical instabilities, resulting

from the absence of an optimum. Choosing the right tensor format representing the refined base measure guarantees the existence of an optimum.

Further research might derive a relation between tensor format complexity [21] to notions of facet complexity [37]. The complexity of faces from randomly drawn polytopes is further interesting, especially in the asymptotic limit of large statistics [41, 42].

References

- [1] Shannon, C.E.: A Mathematical Theory of Communication. Bell System Technical Journal **27**(3), 379–423 (1948) <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x> . Accessed 2025-04-04
- [2] Koller, D., Friedman, N.: Probabilistic Graphical Models: Principles and Techniques, 1. edition edn. The MIT Press, Cambridge, Mass. (2009)
- [3] Yoon, S., Hwang, H., Kwon, D., Noh, Y.-K., Park, F.C.: Maximum entropy inverse reinforcement learning of diffusion models with energy-based models. In: Proceedings of the 38th International Conference on Neural Information Processing Systems. NIPS '24, vol. 37, pp. 24601–24624. Curran Associates Inc., Red Hook, NY, USA (2024)
- [4] Song, L., Guo, Q., Channa, I.A., Wang, Z.: A Survey of Maximum Entropy-Based Inverse Reinforcement Learning: Methods and Applications. Symmetry **17**(10), 1632 (2025) <https://doi.org/10.3390/sym17101632> . Accessed 2025-12-05
- [5] Miotto, M., Monacelli, L.: A new pathway to generative artificial intelligence by minimizing the maximum entropy. arXiv. arXiv:2502.13287 [cs] (2025). <https://doi.org/10.48550/arXiv.2502.13287> . <http://arxiv.org/abs/2502.13287> Accessed 2025-12-05
- [6] Rathnakumar, R., Lu, X., Huang, J., Yan, H., Liu, Y.: Bayesian Entropy Neural Networks for physics-aware prediction. Reliability Engineering & System Safety **266**, 111575 (2026) <https://doi.org/10.1016/j.ress.2025.111575> . Accessed 2025-12-05
- [7] Barndorff-Nielsen, O.E.: Information and Exponential Families: In Statistical Theory. J. Wiley, ??? (1978). Google-Books-ID: Y5muvwEACAAJ
- [8] Csiszar, I., Matus, F.: Closures of exponential families. The Annals of Probability **33**(2) (2005) <https://doi.org/10.1214/009117904000000766> . arXiv:math/0503653. Accessed 2026-07-14
- [9] Malagò, L., Pistone, G.: A note on the border of an exponential family. arXiv. arXiv:1012.0637 [math.ST] (2010). <https://doi.org/10.48550/arXiv.1012.0637> . <http://arxiv.org/abs/1012.0637> Accessed 2026-07-15

- [10] Sottile, F.: Toric ideals, real toric varieties, and the algebraic moment map. arXiv. arXiv:math/0212044 (2008). <https://doi.org/10.48550/arXiv.math/0212044> . <http://arxiv.org/abs/math/0212044> Accessed 2026-07-14
- [11] Straszak, D., Vishnoi, N.K.: Computing Maximum Entropy Distributions Everywhere. ArXiv (2017). Accessed 2026-07-14
- [12] Straszak, D., Vishnoi, N.K.: Maximum Entropy Distributions: Bit Complexity and Stability. In: Proceedings of the Thirty-Second Conference on Learning Theory, pp. 2861–2891. PMLR, ??? (2019). <https://proceedings.mlr.press/v99/straszak19a.html> Accessed 2026-07-14
- [13] De Raedt, L., Frasconi, P., Kersting, K., Muggleton, S. (eds.): Probabilistic Inductive Logic Programming. Lecture Notes in Computer Science, vol. 4911. Springer, Berlin, Heidelberg (2008). <https://doi.org/10.1007/978-3-540-78652-8> . <http://link.springer.com/10.1007/978-3-540-78652-8> Accessed 2026-07-27
- [14] Getoor, L., Taskar, B.: Introduction to Statistical Relational Learning. MIT Press, ??? (2019)
- [15] Sarker, M.K., Zhou, L., Eberhart, A., Hitzler, P.: Neuro-symbolic artificial intelligence: Current trends. AI Communications **34**(3), 197–209 (2022) <https://doi.org/10.3233/AIC-210084> . Accessed 2024-04-17
- [16] Goessmann, A., Schütte, J., Fröhlich, M., Eigel, M.: A tensor network formalism for neuro-symbolic AI. arXiv. arXiv:2601.15442 [cs] (2026). <https://doi.org/10.48550/arXiv.2601.15442> . <http://arxiv.org/abs/2601.15442> Accessed 2026-01-27
- [17] Goessmann, A.: The Tensor-Network Approach to Efficient and Explainable AI (2025). <https://github.com/EnexaProject/enexa-tensor-reasoning-documentation/> Accessed 2025-12-08
- [18] White, S.R.: Density-matrix algorithms for quantum renormalization groups. Physical Review B **48**(14), 10345–10356 (1993) <https://doi.org/10.1103/PhysRevB.48.10345> . Accessed 2025-12-08
- [19] Bellman, R.E.: Adaptive Control Processes. Princeton University Press, New Jersey (1961)
- [20] Hackbusch, W., Kühn, S.: A New Scheme for the Tensor Representation. Journal of Fourier Analysis and Applications **15**(5), 706–722 (2009)
- [21] Hackbusch, W.: Tensor Spaces and Numerical Tensor Calculus. Springer Series in Computational Mathematics. Springer, Berlin Heidelberg (2012). <https://doi.org/10.1007/978-3-642-28027-6>
- [22] Hitchcock, F.L.: The Expression of a Tensor or a Polyadic as a Sum of Products.

- Journal of Mathematics and Physics **6**(1-4), 164–189 (1927) <https://doi.org/10.1002/sapm192761164>
- [23] Beylkin, G., Mohlenkamp, M.J.: Algorithms for Numerical Analysis in High Dimensions. *SIAM Journal on Scientific Computing* **26**(6), 2133–2159 (2005) <https://doi.org/10.1137/040604959>
 - [24] Oseledets, I.V.: Tensor-Train Decomposition. *SIAM Journal on Scientific Computing* **33**(5), 2295–2317 (2011) <https://doi.org/10.1137/090752286> . Accessed 2025-12-08
 - [25] Eigel, M., Schneider, R., Trunschke, P., Wolf, S.: Variational Monte Carlo—bridging concepts of machine learning and high-dimensional partial differential equations. *Advances in Computational Mathematics* **45**(5), 2503–2532 (2019) <https://doi.org/10.1007/s10444-019-09723-8> . Accessed 2026-01-14
 - [26] Eigel, M., Marschall, M., Pfeffer, M., Schneider, R.: Adaptive stochastic Galerkin FEM for lognormal coefficients in hierarchical tensor representations. *Numerische Mathematik* **145**(3), 655–692 (2020) <https://doi.org/10.1007/s00211-020-01123-1> . Accessed 2026-01-15
 - [27] Sander, A., Fröhlich, M., Ali, M., Eigel, M., Eisert, J., Hintermüller, M., Mendl, C.B., Milbradt, R.M., Wille, R.: Quantum circuit simulation with a local time-dependent variational principle. *arXiv*. arXiv:2508.10096 [quant-ph] (2025). <https://doi.org/10.48550/arXiv.2508.10096> . <http://arxiv.org/abs/2508.10096> Accessed 2025-12-15
 - [28] Badreddine, S., Garcez, A., Serafini, L., Spranger, M.: Logic Tensor Networks. *Artificial Intelligence* **303**, 103649 (2022) <https://doi.org/10.1016/j.artint.2021.103649> . Accessed 2023-11-20
 - [29] Domingos, P.: Tensor Logic: The Language of AI. *arXiv*. arXiv:2510.12269 [cs] (2025). <https://doi.org/10.48550/arXiv.2510.12269> . <http://arxiv.org/abs/2510.12269> Accessed 2025-11-12
 - [30] Ali, A.M.: Explicit Solution Equation for Every Combinatorial Problem via Tensor Networks: MeLoCoToN. *arXiv* (2025). <https://doi.org/10.48550/arXiv.2502.05981> . <http://arxiv.org/abs/2502.05981> Accessed 2025-08-28
 - [31] Ziegler, G.M.: *Lectures on Polytopes*, 1995th edition edn. Springer, New York (2013)
 - [32] Brown, L.D.: *Fundamentals of Statistical Exponential Families*. Institute of mathematical Statistics, Hayward, Calif (1987)
 - [33] Wainwright, M.J., Jordan, M.I.: *Graphical Models, Exponential Families, and Variational Inference*. Now Publishers Inc, ??? (2008)

- [34] Ziegler, G.M.: Lectures on 0/1-Polytopes. In: Kalai, G., Ziegler, G.M. (eds.) Polytopes — Combinatorics and Computation, pp. 1–41. Birkhäuser, Basel (2000). https://doi.org/10.1007/978-3-0348-8438-9_1 . https://doi.org/10.1007/978-3-0348-8438-9_1 Accessed 2025-03-04
- [35] Russell, S., Norvig, P.: Artificial Intelligence: A Modern Approach, Global Edition: A Modern Approach, Global Edition, 4th edn. Pearson, Boston (2021)
- [36] Gallager, R.G.: Low-Density Parity-Check Codes. The MIT Press, ??? (1963). <https://doi.org/10.7551/mitpress/4347.001.0001> . <https://direct.mit.edu/books/monograph/3867/Low-Density-Parity-Check-Codes> Accessed 2025-12-05
- [37] Grötschel, M., Lovász, L., Schrijver, A.: Geometric Algorithms and Combinatorial Optimization. Algorithms and Combinatorics, vol. 2. Springer, Berlin, Heidelberg (1993). <https://doi.org/10.1007/978-3-642-78240-4> . <http://link.springer.com/10.1007/978-3-642-78240-4> Accessed 2025-12-05
- [38] Jordan, M.I. (ed.): Learning in Graphical Models. Springer, Dordrecht (1998). <https://doi.org/10.1007/978-94-011-5014-9> . <http://link.springer.com/10.1007/978-94-011-5014-9> Accessed 2026-07-08
- [39] Darroch, J.N., Ratcliff, D.: Generalized Iterative Scaling for Log-Linear Models. The Annals of Mathematical Statistics **43**(5), 1470–1480 (1972). Accessed 2026-07-08
- [40] Csiszar, I.: A Geometric Interpretation of Darroch and Ratcliff’s Generalized Iterative Scaling. The Annals of Statistics **17**(3), 1409–1413 (1989) <https://doi.org/10.1214/aos/1176347279> . Accessed 2026-07-08
- [41] Vershynin, R.: High-Dimensional Probability: An Introduction with Applications in Data Science, 1st edition edn. Cambridge University Press, New York, NY (2018)
- [42] Wainwright, M.J.: High-Dimensional Statistics: A Non-Asymptotic Viewpoint. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge (2019). <https://doi.org/10.1017/9781108627771>